Lab 14: Balmer Series & Rydberg Constant

Activity #1: Measuring Lines per Unit Length of the Diffraction Gratting (Mercury Source)

Setup Parameters		
L (grating to ruler)	0.3	m
Diffraction order (m)	1	

NIST Mercury Wavelengths (Source: https://physics.nist.gov/PhysRefData/Handbook/Tables/mercurytable2.htm)

Color	**λ (m) [NIST]**	**x_right (m)**	**x_left (m)**	**x_avg (m)**	**θ (rad)**	**d (m)**	**N (lines/mm)**
Red	0.000000691	0.26	0.26	0.26	0.7140906986	1.06E-06	9.48E+02
Green/Cyan	0.00000054€	0.195	0.21	0.2025	0.5937496667	9.76E-07	1.02E+03
Violet	4.36E-07	0.144	0.145	0.1445	0.4488736672	1.00E-06	9.95E+02
					Average:	1.01E-06	9.89E+02

Formula used: $d = (m \times \lambda) / \sin(\theta)$ where $\theta = \arctan(x / L)$

Activity #2 — Verify Balmer's Formula & Measure Rydberg Constant (R)

Setup Parameters		
d_avg from Activity #1:	1.01E-06	m
Diffraction order (m):	1	

Color	**x_right (m)**	**x_left (m)**	**x_avg (m)**	**θ (rad)**	**λ_exp (m)**	**n**	**1/λ_exp (m^-1**	**R_measured (m^-1)**
Red	0.27	0.275	0.2725	0.7374001:	6.80E-07	3	1.47E+06	1.06E+07
Green	0.19	0.19	0.19	0.5645693!	5.41E-07	4	1.85E+06	9.85E+06
Violet	0.152	0.155	0.1535	0.4729373:	4.61E-07	5	2.17E+06	1.03E+07
							Average R:	1.03E+07

Formula used: $\lambda = d \cdot \sin(\theta)/m$ $R = (1/\lambda) / (1/2^2 - 1/n^2)$

Expected R (eClass):	1.10E+07
R_exp	1.03E+07
% Error	6.52